

We shall deal with positive form as linear operator at first, and we will generalize.

Def. A linear operator  $T: V \rightarrow V$  on a fin. dim inn. prod space  $V$  is called positive if:

$T$  is Hermitian and  $\langle T\alpha, \alpha \rangle \geq 0$  for all  $\alpha \neq 0$

Prop. Let  $T, U$  be Hermitian on fin. dim inn. prod space  $V$ .

And,  $A = [T]_{\beta}$ , where  $\beta$  is any orthonormal basis for  $V$ .

1).  $T$  is positive if and only if all eigenvalues of  $T$  are positive

2).  $T$  is positive if and only if  $\sum_{i=1}^n A_{ii} a_i \bar{a}_i \geq 0$  for all nonzero tuple  $(a_1, \dots, a_n)$ .

3).  $T$  is positive if and only if

$$A = B^* B \text{ for some } B.$$

Proof. Denote  $\beta = \{\alpha_1, \dots, \alpha_n\}$ , Let  $B = [I]_{\beta}^{\{\alpha_1, \dots, \alpha_n\}} = (\alpha_1 \ \alpha_2 \ \dots \ \alpha_n)$ .

And, let  $\beta' = \{\beta_1, \dots, \beta_n\}$  be an orthonormal basis for  $V$  consisting eigenvectors of  $T$ .

1). If  $T$  is positive, then  $\langle T(\beta_i), \beta_i \rangle = \langle C_i \beta_i, \beta_i \rangle = C_i \geq 0$ , where  $C_i$  is eigenvalue.

Conversely, given  $X \in V$ ,  $X = x_1 \beta_1 + \dots + x_n \beta_n = \langle X, \beta_1 \rangle \cdot \beta_1 + \dots + \langle X, \beta_n \rangle \cdot \beta_n$ , so

$$\begin{aligned} \langle T(X), X \rangle &= \langle \langle X, \beta_1 \rangle \cdot C_1 \beta_1 + \dots + \langle X, \beta_n \rangle \cdot C_n \beta_n, \langle X, \beta_1 \rangle \cdot \beta_1 + \dots + \langle X, \beta_n \rangle \cdot \beta_n \rangle \\ &= \sum_{i=1}^n C_i \cdot |\langle X, \beta_i \rangle|^2 \geq 0 \text{ if } X \neq 0 \text{ and } C_i \geq 0. \end{aligned}$$

2). Given  $X \in V$  st  $[X]_{\beta} = (a_1, \dots, a_n)$ ,

$$\langle T(X), X \rangle = \left\langle \sum_{j=1}^n a_j T(\alpha_j), \sum_{i=1}^n a_i \alpha_i \right\rangle = \sum_{j=1}^n \sum_{i=1}^n \langle T(\alpha_j), \alpha_i \rangle \cdot a_j \cdot \bar{a}_i, \text{ so proved.}$$

(+Recall:  $A_{ij} = \langle T(\alpha_j), \alpha_i \rangle$ ).

3). If  $A = B^* B$ , then  $\langle B^* B x, x \rangle = \langle Bx, Bx \rangle = \|Bx\|^2 \geq 0$ . Conversely,

since  $[T]_{\beta} = D$  is diagonal with positive values, where

$$D = \begin{pmatrix} C_1 & & 0 \\ & C_2 & \\ 0 & & C_n \end{pmatrix} \text{ } C_i \text{'s are eigenvalues of } T$$

so we can define  $\sqrt{D} = \begin{pmatrix} \sqrt{C_1} & & 0 \\ & \sqrt{C_2} & \\ 0 & & \sqrt{C_n} \end{pmatrix}$ . Now, for  $P = [I]_{\beta}^{\beta}$ ,

$$A = P^{-1} D P = P^{-1} \cdot \sqrt{D} \cdot \sqrt{D} \cdot P = (\sqrt{D} P)^* (\sqrt{D} P), \text{ since } P \text{ is unitary, } \sqrt{D} \text{ is symmetric.}$$

And, there are more interesting problems,

4). If  $T$  and  $U$  are positive s.t.  $T^2 = U^2$ , then  $T = U$ .

5). If  $T$  and  $U$  are positive s.t.  $TU = UT$ , then  $TU$  is positive.

Proof. Let  $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$  be an orthonormal basis for  $V$  consisting of eigenvectors of  $T$ . Let  $C_1, \dots, C_n$  be eigenvalues corresponding to  $\alpha_1, \dots, \alpha_n$ , respectively. Setting for each  $1 \leq i \leq n$ ,

$$T^2(\alpha_i) = C_i^2 \cdot \alpha_i = U^2(\alpha_i).$$

$$\text{And, } C_i^2 \alpha_i = U^2(\alpha_i) \Leftrightarrow (U^2 - C_i I)(\alpha_i) = 0.$$

$$\Leftrightarrow (U + C_i I) \circ (U - C_i I)(\alpha_i) = 0.$$

If  $V = (U - C_i I)(\alpha_i) \neq 0$ , then

$$U(V) = -C_i V$$

So  $V$  is an eigenvector of  $U$  with negative eigenvalue.

It is contradiction to  $U$  is positive, so  $U(\alpha_i) = C_i \alpha_i$  for all  $1 \leq i \leq n$ .

The linear map is determined by maps basis to some vectors, so  $T = U$  (4)

Since  $T$  and  $U$  commute and both are Hermitian,

$$(TU)^* = U^* T^* = UT = TU$$

So  $TU$  is Hermitian. For the above  $\mathcal{B}$ , define a linear map as

$$\sqrt{T}(\alpha_i) = \sqrt{C_i} \cdot \alpha_i.$$

Then, for each  $1 \leq i \leq n$ ,

$$U\sqrt{T}(\alpha_i) = U(\sqrt{C_i} \alpha_i) = \frac{1}{\sqrt{C_i}} \cdot U(C_i \alpha_i) = \frac{1}{\sqrt{C_i}} \cdot U(T(\alpha_i)) = \frac{1}{\sqrt{C_i}} \cdot TU(\alpha_i)$$



Def. A form  $f$  on an inn. prod space  $V$  is called non-negative if:

$f$  is Hermitian and  $f(\alpha, \alpha) \geq 0$  for all  $\alpha \in V$ .

and, if  $f(\alpha, \alpha) > 0, \alpha \neq 0$ , it is called positive,

Exmple. Let  $f$  be a form on fin. dim inn. prod space  $V$ , let  $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$  be an ordered basis.

Let  $A$  be a matrix of  $f$  in  $\mathcal{B}$ . That is,  $A_{jk} = f(\alpha_k, \alpha_j)$

If  $\alpha = x_1 \alpha_1 + \dots + x_n \alpha_n$ , then

$$f(\alpha, \alpha) = f\left(\sum_j x_j \alpha_j, \sum_k x_k \alpha_k\right) = \sum_j \sum_k x_j \overline{x_k} \cdot f(\alpha_j, \alpha_k) = \sum_j \sum_k A_{kj} \cdot x_j \cdot \overline{x_k}.$$

So,  $f$  is non-negative if and only if  $A = A^*$  and  $\sum_j \sum_k A_{kj} x_j \overline{x_k} \geq 0$ , for all  $x_j, x_k$

Equivalently,  $f$  is non-negative if and only if

$$\mathcal{Q}: F^n \times F^n \rightarrow F: (X, Y) \mapsto Y^* A X$$

is non-negative form,  $F = \mathbb{C}$  or  $\mathbb{R}$ ,

Thm. Let  $F = \mathbb{C}$  or  $\mathbb{R}$ , and  $A \in M_{\text{sym}}(F)$ .

A function  $\mathcal{Q}: F^n \times F^n \rightarrow F: (X, Y) \mapsto Y^* A X$

is a positive form if and only if  $A = P^* P$  for some invertible  $P \in M_{\text{inn}}(F)$ .

Proof Suppose that  $\mathcal{Q}$  is positive form. Then, it is an inner product on  $F^n$ .

So there is an orthonormal basis  $\{\alpha_1, \dots, \alpha_n\}$  for  $F^n$ , satisfying

$$\mathcal{Q}(\alpha_i, \alpha_j) = \alpha_j^* A \alpha_i = \delta_{i,j}.$$

so, take  $P = (\alpha_1 \dots \alpha_n)^{-1}$ , then  $P A P^* = I$ , or  $A = P^* P$ .

Conversely, if  $A = P^* P$  for some invertible  $P$ , then  $A^* = (P^* P)^* = P^* P = A$  and

$$X^* A X = X^* P^* P X = (P X)^* \cdot (P X) > 0 \text{ if } X \neq 0.$$

So proved.

Def. Let  $A \in M_{n \times n}(F)$ . The principal minors of  $A$  are:

$$\Delta_k(A) \stackrel{\text{def}}{=} \det \begin{bmatrix} A_{11} & \dots & A_{1k} \\ \vdots & \ddots & \vdots \\ A_{k1} & \dots & A_{kk} \end{bmatrix} \quad 1 \leq k \leq n$$

Lemma. Let  $A \in M_{n \times n}(F)$  be an 'invertible'. TFAE:

- There is a uni-upper triangle matrix  $P$  such that  $AP$  is lower triangular.
- The principal minors of  $A$  are all different from 0.

Proof. Note:  $B$  is upper triangle iff  $B_{jk} = 0$  if  $j > k$ .

Suppose that a) holds.

Denote  $A_1, \dots, A_n$  are columns of  $A$ , and  $B_1, \dots, B_n$  are column of  $B = AP$ .

Then,  $B_1 = A_1$  by  $P$  is uni-upper, and

$$B_r = A_r + \sum_{j=1}^{r-1} p_{jr} \cdot A_j$$

(observe that

$$\begin{aligned} (B_1 \dots B_n) &= (A_1 \dots A_n) \cdot \left( I + \begin{pmatrix} 0 & & p_{1k} \\ & \ddots & \\ 0 & & 0 \end{pmatrix} \right) \quad (p_{jk} = 0 \text{ if } j > k) \\ &= (A_1 \dots A_n) + \underbrace{\left( \sum_{j=1}^{r-1} p_{jr} \cdot A_j \right)}_{r\text{th column}} \end{aligned}$$

Therefore, for each  $1 \leq k \leq n$ ,

$$\begin{bmatrix} B_{11} & \dots & B_{1k} \\ \vdots & \ddots & \vdots \\ B_{k1} & \dots & B_{kk} \end{bmatrix} \text{ is obtained by adding to the } r\text{th column of}$$

$$\begin{bmatrix} A_{11} & \dots & A_{1k} \\ \vdots & \ddots & \vdots \\ A_{k1} & \dots & A_{kk} \end{bmatrix} \text{ as a linear combination of its other columns.}$$

This operation cannot change the determinant, so,  $\Delta_k(A) = \Delta_k(B)$ .

And, since  $A, P$  are invertible, so is  $B$ , thus  $\Delta_n(B) = \det B \neq 0$ .

Now,  $\Delta_k(B) = B_{11} \dots B_{kk} \neq 0$ , since it is triangular.